OJASH GUPTA

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OBJECTIVE

Motivated 3rd-year B.Tech student in Artificial Intelligence at SVNIT, Surat, with strong interest in Machine Learning, software development, and problem solving using C++ and data structures. Seeking an internship opportunity to build reliable, efficient, and scalable software while gaining practical experience in performance optimization, Linux systems, and real-world engineering workflows.

EDUCATION

B.Tech - Artificial Intelligence, SVNIT, Surat | 2024 - Present | GPA: 8.47/10

Senior Secondary (XII), Kendriya Vidyalaya, Bhusawal | 2017 - 2023 | Score: 96.2%

SKILLS

Languages: C++, Python

Core: Data Structures & Algorithms, Machine Learning Fundamentals, Regression Models, RAG, LLM Security Basics, Robotics

Tools/Frameworks: LangChain, FAISS, Streamlit, FastAPI, React, MongoDB, REST APIs, JWT, Git, Microcontrollers

Platforms: GitHub, Unix/Linux, VS Code

PROJECTS

AI RAG PDF Chatbot | May 2026 | Python, LangChain, FAISS, Streamlit, Embeddings, Vector Search | GitHub

- Built a Retrieval-Augmented Generation chatbot that allows users to upload PDFs and ask context-aware questions from document content.
- Implemented PDF loading, text chunking, embedding generation, FAISS-based vector similarity search, and LLM response generation.
- Improved answer relevance by tuning chunk size, overlap, and top-k retrieval so the model receives stronger document context.
- Reduced hallucination risk by grounding answers on retrieved chunks and adding fallback prompts for missing or unclear context.

Bloom Branding - Full-Stack CMS Platform (GWOC 2025) | Feb 2026 | GitHub

- Built a full-stack monorepo with Node.js/Express backend and multiple React/Next.js frontends for content management workflows.
- Implemented REST APIs, JWT authentication, MongoDB models, and Cloudinary-based media handling for secure admin operations.
- Developed admin dashboard and customer-facing UI using TypeScript and Tailwind CSS, improving maintainability and user experience.

LLM Prompt Injection Protection Extension | Feb 2026 | ML, Regression Models, LLM Security, Python | GitHub

- Designed a security extension for LLMs to detect and mitigate prompt injection attacks using regression-based classification models.
- Classified prompts into safe, suspicious, and malicious categories; allowed safe inputs, analyzed suspicious inputs, and blocked unsafe prompts.
- Monitored LLM outputs for potential sensitive information leakage to support safer and more reliable AI interactions.

Line Follower Robot

- Designed an autonomous robot using IR sensors and microcontroller-based control logic for line tracking and navigation.
- Improved movement stability by testing sensor thresholds and refining control behavior for turns and path correction.

KMeansAccel - Parallel K-Means Benchmark | Jul 2026 | C++, OpenMP, CUDA | GitHub

- Benchmarked sequential, OpenMP, and CUDA K-means on the 581K-row UCI Covertype dataset to compare scalability.
- Added thread-local reductions, CUDA kernels, result validation, automated timing, and speedup visualizations.

AI FOCUS FOR INTERNSHIP

- Interested in MCP-style tool specs, prompt versioning, agentic slot-booking flows, graceful agent handoffs, and structured evaluation of healthcare chatbots.
- Comfortable iterating on prompts and retrieval logic to improve answer quality, reduce vague responses, and handle missing or unclear user inputs.

ACHIEVEMENTS | Certificates

- Top 6 Finalist - Line Follower Robot Challenge (2025).
- Winner - Coding Contest, Nexus Club Mentorship Program (2025).
- Finalist - ECHELON National Level AI Cybersecurity Hackathon.

ACTIVITIES

- Active on LeetCode and Codeforces(Pupil,1300+)